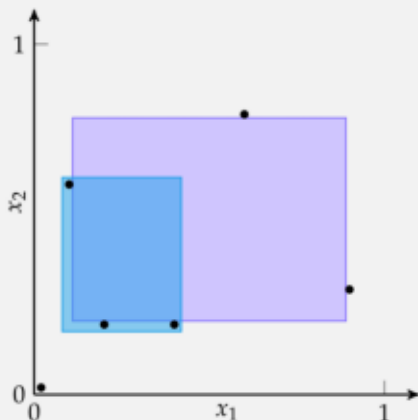


Consider the set:

$$X = \left\{ \left[\frac{1}{5}, \frac{1}{5} \right], \left[\frac{2}{5}, \frac{1}{5} \right], \left[\frac{1}{10}, \frac{3}{5} \right], \left[\frac{9}{10}, \frac{3}{10} \right], \left[\frac{1}{50}, \frac{1}{50} \right], \left[\frac{3}{5}, \frac{4}{5} \right] \right\}$$

The discrepancy of X with respect to the unit square is determined by a rectangular subset H that either has very small area but contains very many points or has very large area and contains very few points.



The blue rectangle, $x_1 \in \left[\frac{1}{10}, \frac{2}{5} \right]$, $x_2 \in \left[\frac{1}{5}, \frac{3}{5} \right]$, has a volume of 0.12 and contains 3 points. Its corresponding discrepancy measure is thus 0.38.

The purple rectangle, $x_1 \in \left[\frac{1}{10} + \epsilon, \frac{9}{10} - \epsilon \right]$, $x_2 \in \left[\frac{1}{5} + \epsilon, \frac{4}{5} - \epsilon \right]$, produces an even higher discrepancy. As ϵ approaches zero, the volume and the discrepancy approach 0.48 because the rectangle contains no points. Note that the limit was required, reflecting the need to use a supremum in the definition

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